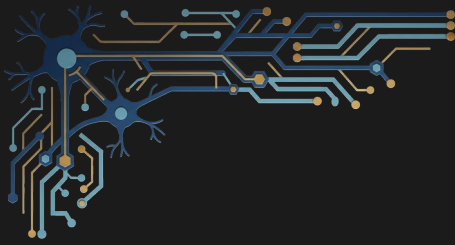




Neurobotics™

The Future is Here. Real. Now.

Neurobotics Medical Horizons



Neurobotics Medical Horizons

The evolution from edge-compute robotics to medical bio-cybernetics demands extreme hardware miniaturization. This streamlined public briefing outlines the clinical capabilities of the architecture while fully protecting proprietary silicon mechanics.

The Silicon Horizon

- The architecture is transitioning from a 4"x4" sub-15W printed circuit board (SBUD-SB2) to a custom System-on-Chip (SoC) and Microcontroller Unit (MCU).
- This lithographic compression reduces the physical footprint to millimeters and scales power consumption down to milliwatt thresholds.
- Ultra-low-power edge compute enables long-term battery operation and passive bio-galvanic energy harvesting for minimally invasive implants.

Clinical Bio-Cybernetics

- **Spinal Cord Bridging (Tier 1):** Millimetric MCUs can bridge severed spinal regions to read biological nerve impulses and execute real-time motor outputs without external cloud processing.
- **Neurological Stabilization (Tier 2):** Bio-implants monitor physiological drives to detect electrical precursors to seizures or Parkinsonian tremors, deploying real-time micro-stimulation to stabilize the patient.
- **Neuroprosthetic Shielding (Board Four):** Optoisolation protects delicate neural interfaces from high-voltage motor feedback spikes, ensuring signal integrity across the brain-prosthetic barrier.
- **Deterministic Motor Control (Tiers 4-5):** Translates physical pressure into sensory vectors while providing stable kinesthetic gait control.

The Silicon Twin Model

- The 7-Tier S-M-R Matrix operates as a deterministic digital twin of biological neurology by mapping exact anatomical pipelines into a silicon framework.

- Pharmaceutical and academic labs can evaluate brain-computer interfaces (BCIs), neural stress responses, and behavioral valence loops entirely inside a simulated environment.
- This architecture guarantees identical, mathematically verifiable repeatability, effectively eradicating the need for traumatic live animal testing in preclinical trials.

Dual-Licensing Framework

- **Open Science (AGPL-3.0):** University neuro-engineering facilities and non-profit hospitals receive open access to lower-nervous-system schematics for cruelty-free academic validation.
- **Commercial Deployment:** Medical device manufacturers and defense integrators utilize Master Commercial Licenses to build proprietary, closed-loop bio-implants directly upon the Neurobotics IP foundation.

